LESSON 01 / 12

Differentiate the relation with respect to time

If $y = f(x)$ and x changes with time t, the chain rule gives

$$\frac{dy}{dt} = f'(x)\frac{dx}{dt}.$$

The average rate over a time interval is a secant slope; the instantaneous rate is its limit. Units provide a check: if r is in cm and t in seconds, then $d(\pi r^2)/dt$ is in cm^2/s.

WORKED EXAMPLE

Circle, cube and balloon: choose the geometric relation first

For a circle with $dr/dt = 2$ cm/s,

$$A = \pi r^2, \qquad \frac{dA}{dt} = 2\pi r\frac{dr}{dt} = 4\pi r.$$

At $r = 5$, the area grows at 20π cm^2/s.

For a cube with $ds/dt = 3$ cm/min,

$$V = s^3, \qquad \frac{dV}{dt} = 3s^2\frac{ds}{dt},$$

so $s = 10$ gives 900 cm^3/min.

For a spherical balloon with $dV/dt = 100$ cm^3/s,

$$V = \frac{4}{3}\pi r^3, \qquad \frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}.$$

Thus at $r = 10$, $dr/dt = 1/(4\pi)$ cm/s, approximately 0.0796 cm/s.

WORKED EXAMPLE

The 13 m ladder: a negative rate means descent

Let x be the foot's distance from the wall and y the top's height. Then

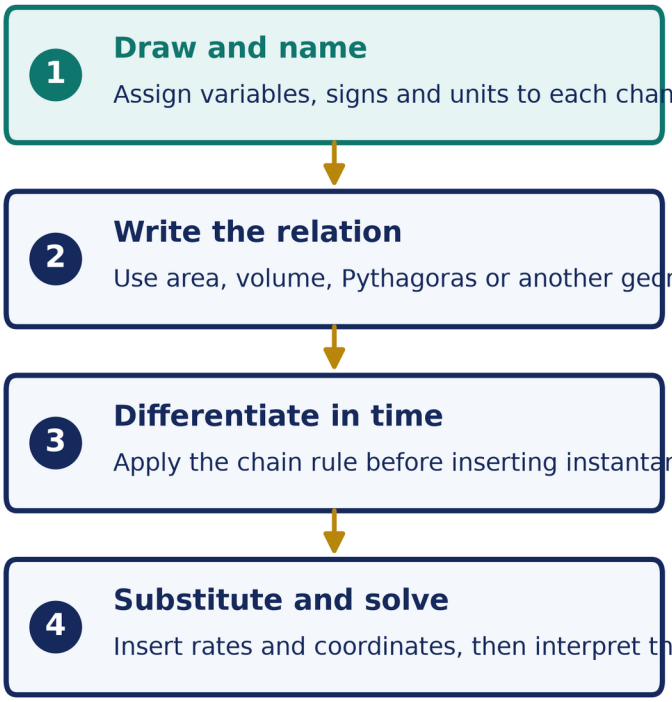
$$x^2 + y^2 = 169 \quad\Longrightarrow\quad 2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0.$$

When $x = 5$, $y = 12$ and $dx/dt = 1/2$, so

$$\frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{5}{12}\cdot\frac{1}{2} = -\frac{5}{24}\ \text{m/s}.$$

The top descends at $5/24$ m/s, approximately 0.2083 m/s.

A related-rates calculation keeps every variable alive



READ THE FIGURE

Substituting a length before differentiating would freeze that length and incorrectly make its rate zero.

COMMON TRAP

Differentiate first, substitute last

A numerical value such as $x = 5$ describes one instant. It does not say that x is constant while the ladder moves. Keep x and y variable until after the time derivative has been formed.

LESSON 02 / 12

Point, slope, then perpendicular slope

At $(a, f(a))$ with finite derivative $m = f'(a)$, the tangent is

$$y - f(a) = m(x - a).$$

If $m \neq 0$, the normal has slope $-1/m$. If $m = 0$, the tangent is horizontal and the normal is vertical. A vertical tangent such as $x = 0$ for $y = x^{1/3}$ needs a geometric statement, not substitution into a finite-slope formula.

WORKED EXAMPLE

A cubic tangent and normal at x=2

For $f(x) = x^3 - 3x$, $f(2) = 2$ and $f'(2) = 3(4) - 3 = 9$. Thus

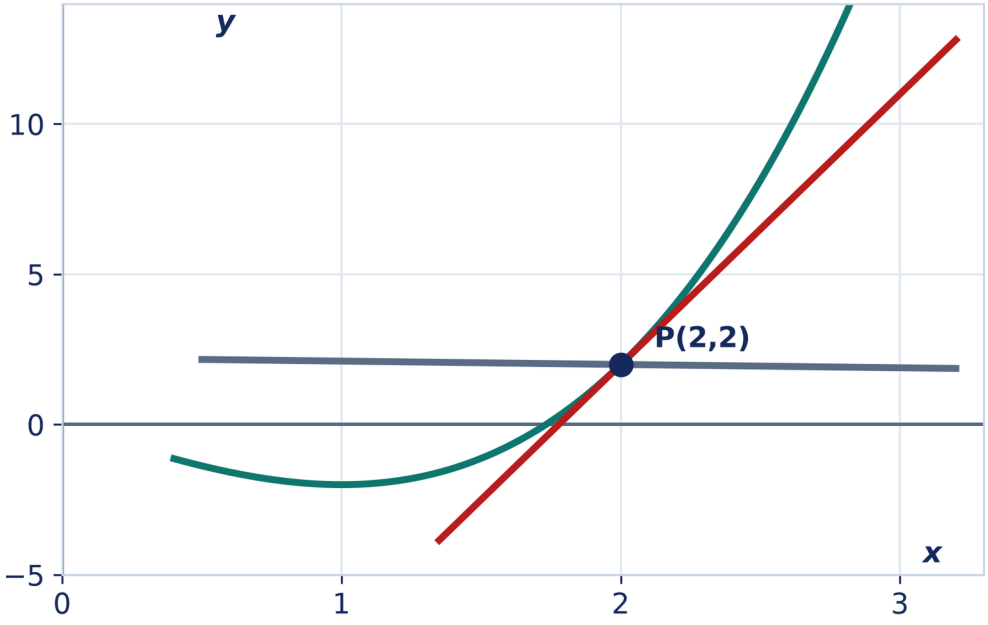
$$\text{tangent: } y - 2 = 9(x - 2) \quad \Longrightarrow \quad y = 9x - 16,$$

and

$$\text{normal: } y - 2 = -\tfrac{1}{9}(x - 2) \quad \Longrightarrow \quad x + 9y = 20.$$

For comparison, the normal to $y = x^2$ at $(1, 1)$ has slope $-1/2$, hence equation $x + 2y = 3$.

The cubic tangent and normal are perpendicular at (2,2)



READ THE FIGURE

The teal cubic has tangent y=9x-16 in red and normal x+9y=20 in slate. Their slopes 9 and -1/9 multiply to -1.

Parallel chords and angles use tangent slopes

The chord of $y = x^2$ joining $(1, 1)$ and $(3, 9)$ has slope $(9 - 1)/(3 - 1) = 4$. Solving $2x = 4$ gives the parallel tangent at $(2, 4)$.

For nonvertical tangents of slopes m_1, m_2 with $m_1 m_2 \neq -1$, the acute angle θ between them satisfies

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|.$$

At $(1, 1)$, $y = x^2$ and $y = x^3$ have slopes 2 and 3, so $\tan\theta = 1/7$ and $\theta = \arctan(1/7) \approx 8.13^\circ$. Orthogonal curves instead have $m_1 m_2 = -1$; the angle formula's denominator then vanishes.

WORKED EXAMPLE

Check orthogonality at the actual intersections

For $y^2 = 4x$ and $x^2 + y^2 = 5$, substitution gives $x^2 + 4x - 5 = 0$. The real intersections are $(1, 2)$ and $(1, -2)$. Implicit differentiation gives

$$m_1 = \frac{2}{y}, \qquad m_2 = -\frac{x}{y}, \qquad m_1 m_2 = -\frac{2x}{y^2} = -\frac{1}{2}$$

at either intersection. The curves are therefore **not** orthogonal.

COMMON TRAP

Do not use the angle formula when the tangents are perpendicular

The condition $m_1 m_2 = -1$ is handled separately. Vertical tangents also require a line-angle or implicit-geometric analysis rather than the displayed slope quotient.

LESSON 03 / 12

Derive lengths from the x-axis intersections

At $P=(x_0,y_0)$, suppose $m=f'(x_0)\neq 0$ and $y_0\neq 0$. The tangent meets the x-axis at $T=(x_0-y_0/m,0)$, while the normal meets it at $N=(x_0+my_0,0)$. Horizontal projections give the subtangent and subnormal; the Pythagorean theorem gives the segment lengths.

Tangent and normal lengths come from the same two intersections

Quantity	Formula	Geometric meaning
Subtangent	|y0/m|	horizontal projection
Tangent segment	|y0| sqrt(1+m^2)/|m|	P to tangent intercept
Subnormal	|m y0|	horizontal projection
Normal segment	|y0| sqrt(1+m^2)	P to normal intercept

READ THE FIGURE
These formulas concern segments to the x-axis and require the stated nonzero cases. Horizontal or vertical lines are handled directly.

For $y=x^3$ at $(1,1)$, $m=3$. Hence the subtangent is $1/3$, the subnormal is 3, the tangent segment is $\sqrt{10}/3$, and the normal segment is $\sqrt{10}$.

A differential is the tangent-line increment

For small Δx,

$$f(x+\Delta x)\approx f(x)+f'(x)\Delta x.$$

This is a local approximation, not an exact identity. A useful check is to compare with a direct numerical evaluation and, when possible, estimate the error using the next derivative.

WORKED EXAMPLE

Three standard linear approximations

For $f(x)=\sqrt{x}$ at $x=25$, $f'(25)=1/10$, so

$$\sqrt{25.4}\approx 5+\tfrac{1}{10}(0.4)=5.04.$$

The direct value is $5.039841\ldots$, so the approximation overestimates by about 1.59×10^{-4}.

For $f(x)=x^{10}$ at $x=1$,

$$(1.01)^{10}\approx 1+10(0.01)=1.1,$$

compared with $1.104622\ldots$. For $f(x)=\ln x$ at $x=1$,

$$\ln(1.02)\approx 0+1(0.02)=0.02,$$

compared with $0.019803\ldots$.

COMMON TRAP

A small step does not make the linear answer exact

The approximation omits higher-order terms. In the square-root example, the curve is concave, so its tangent lies above the graph and the estimate is slightly too large.

LESSON 04 / 12

The Mean Value Theorem turns derivative sign into graph direction

If $f' > 0$ throughout an interval, then for $a < b$ in that interval, MVT gives c with

$$f(b) - f(a) = f'(c)(b - a) > 0.$$

Thus f is strictly increasing. Similarly, $f' < 0$ implies strictly decreasing. Positive derivative is sufficient for strict increase, but not necessary: x^3 is strictly increasing on $\mathbb{R}$ even though its derivative is zero at 0.

WORKED EXAMPLE

Factor the derivative before drawing the sign chart

For $f(x) = x^3 - 3x$,

$$f'(x) = 3(x - 1)(x + 1).$$

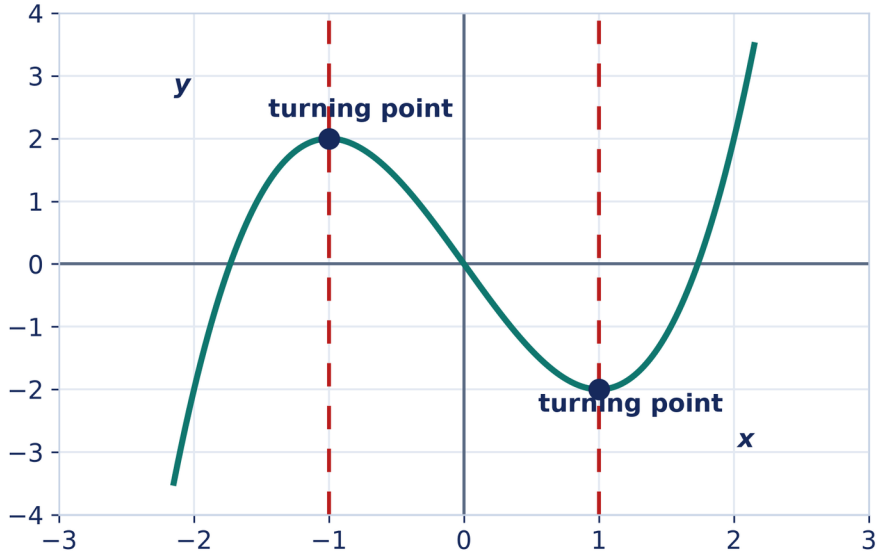
Therefore f increases on $(-\infty, -1)$ and $(1, \infty)$, and decreases on $(-1, 1)$.

For $f(x) = 2x^3 - 9x^2 + 12x$,

$$f'(x) = 6(x - 1)(x - 2),$$

so it increases on $(-\infty, 1)$ and $(2, \infty)$, and decreases on $(1, 2)$.

The derivative sign partitions the cubic into three monotone pieces



READ THE FIGURE

For x^3-3x, the derivative is positive on the outer intervals and negative between -1 and 1. The marked points are where the sign changes.

Monotonicity proves inequalities and global extrema

For $x > 0$, let

$$g(x) = x - 1 - \ln x, \qquad g'(x) = 1 - \tfrac{1}{x}.$$

Then g decreases on $(0, 1)$ and increases on $(1, \infty)$, so its global minimum is $g(1) = 0$. Hence $\ln x \le x - 1$, with equality only at $x = 1$.

Similarly, $F(x) = x - \ln x$ has $F'(x) = 1 - 1/x$, so its minimum value on $x > 0$ is $F(1) = 1$.

COMMON TRAP

A zero at an isolated point does not destroy strict monotonicity

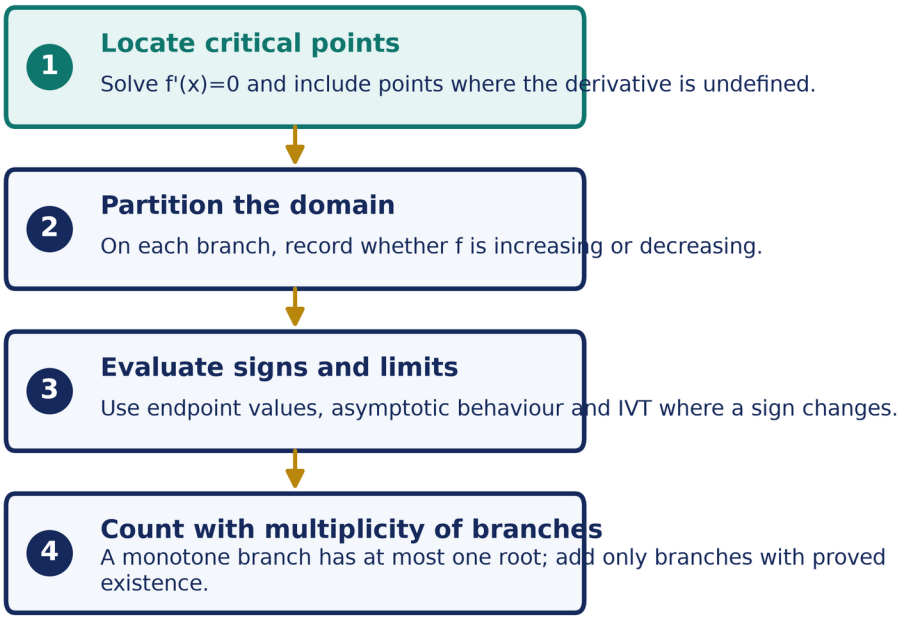
The correct refinement is: if $f' \ge 0$ and f' is not identically zero on any subinterval, then f is strictly increasing. Do not replace this with the false claim that $f'(x) > 0$ at every single point is necessary.

LESSON 05 / 12

Separate existence from uniqueness

The Intermediate Value Theorem supplies a root when a continuous function changes sign. Strict monotonicity on an interval prevents two roots there. First split the domain at turning points and singularities; then apply these two ideas branch by branch.

A root count is an interval-by-interval argument



READ THE FIGURE

A sign change alone does not prove uniqueness, and monotonicity alone does not prove existence.

WORKED EXAMPLE

The cubic x^3-3x+1 has exactly three real roots

For $f(x) = x^3 - 3x + 1$, the critical points are -1 and 1, and

$$f(-2) = -1, \qquad f(-1) = 3, \qquad f(1) = -1, \qquad f(2) = 3.$$

The derivative $3(x^2 - 1)$ makes f strictly monotone on

$$(-\infty, -1), \qquad (-1, 1), \qquad (1, \infty).$$

Its end behaviour is $f(x) \to -\infty$ as $x \to -\infty$ and $f(x) \to \infty$ as $x \to \infty$. Thus IVT gives one root on each branch, and monotonicity rules out a second: exactly three real roots in total.

WORKED EXAMPLE

A line cannot intersect a strictly convex parabola more than twice

For $y = x^2$ and $y = mx + c$, intersections satisfy

$$x^2 - mx - c = 0.$$

This quadratic has two, one or no real solutions according as its discriminant $m^2 + 4c$ is positive, zero or negative. More generally, subtracting the proposed line and counting turning points converts an intersection question into a root count.

COMMON TRAP

Do not infer the number of roots from the degree alone

A cubic has at most three real roots, not necessarily three. The derivative sign chart and endpoint information determine the actual count.

LESSON 06 / 12

A critical point is a candidate, not yet an extremum

A local maximum at an interior point requires nearby values to be no larger; a local minimum requires nearby values to be no smaller. Endpoints are not local extrema in the two-sided interior sense, but they are essential in absolute-extremum problems.

First-derivative test. If f' changes from $+$ to $-$ at c, then f has a local maximum at c; a change from $-$ to $+$ gives a local minimum. If the sign does not change, there is no local extremum.

Second-derivative test. If $f'(c) = 0$ and $f''(c) > 0$, there is a strict local minimum; if $f''(c) < 0$, there is a strict local maximum. When $f''(c) = 0$, the test is inconclusive: compare x^3, x^4 and $-x^4$ at the origin.

WORKED EXAMPLE

Factor the derivative and read both signs

For $f(x) = x^3 - 3x$,

$$f'(x) = 3(x-1)(x+1).$$

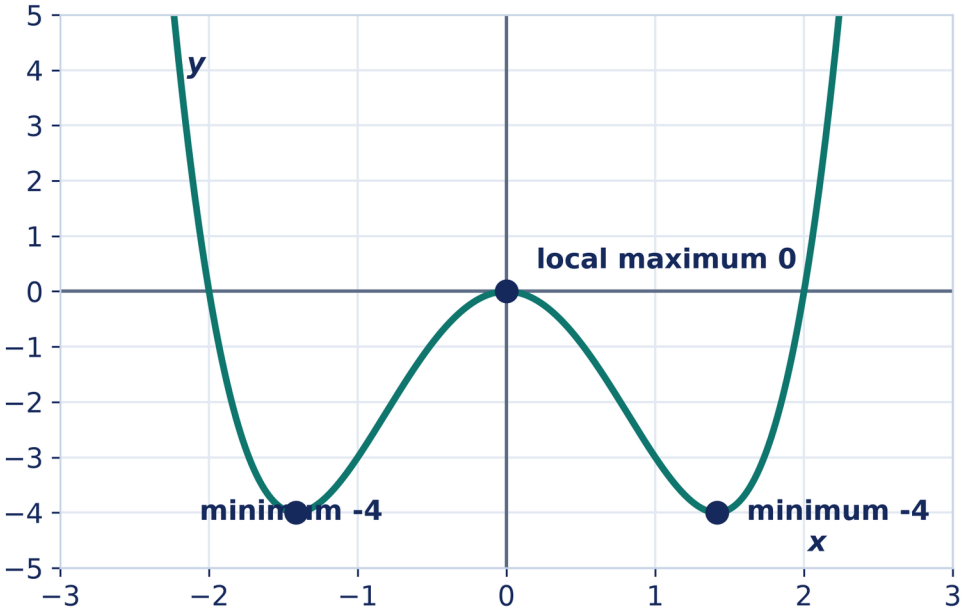
At $x = -1$, the sign changes from positive to negative, so the local maximum is $f(-1) = 2$. At $x = 1$, it changes from negative to positive, so the local minimum is $f(1) = -2$.

For $g(x) = x^3 - 6x^2 + 9x + 2$,

$$g'(x) = 3(x-1)(x-3).$$

Hence g has a local maximum $g(1) = 6$ and a local minimum $g(3) = 2$. The requested local maximum value is therefore 6.

A quartic can have two local minima and one local maximum



READ THE FIGURE

For x^4-4x^2, the derivative 4x(x^2-2) gives critical points -sqrt(2), 0 and sqrt(2).

For this quartic, $h''(x) = 12x^2 - 8$ confirms minima at $\pm\sqrt{2}$ and a maximum at 0. Both minima are also global: $x^4 - 4x^2 = (x^2 - 2)^2 - 4 \geq -4$, with equality at both points.

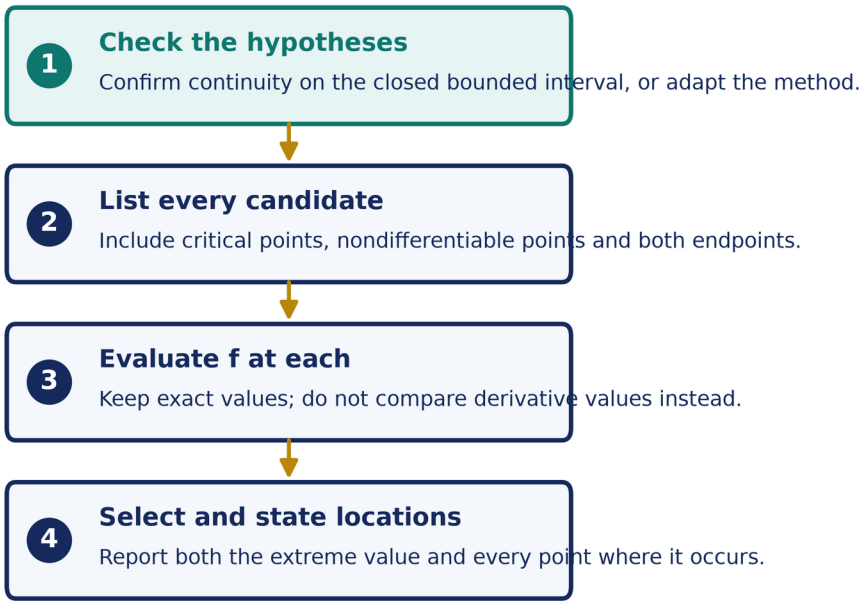
> **COMMON TRAP**
>
> **A point with f'(c)=0 can be none of the three**
>
> The origin for x^3 is stationary but is neither a local maximum nor a local minimum. A sign chart is safer than memorising a label for every stationary point.

LESSON 07 / 12

The Extreme Value Theorem supplies the checklist

A continuous function on a closed bounded interval attains both an absolute maximum and an absolute minimum. Compare the values at every interior stationary point, every point where the derivative fails to exist, and both endpoints.

Absolute extrema are decided by comparing all candidates



READ THE FIGURE
On an open or unbounded interval, values may only approach a supremum or infimum; attainment must be checked separately.

WORKED EXAMPLE

Four candidates determine the extrema on [-2,2]

On $[-2,2]$, for $f(x)=x^3-3x$,

$$f'(x)=3(x-1)(x+1).$$

Both -1 and 1 are interior critical points. The candidate values are

$$f(-2)=-2,\qquad f(-1)=2,\qquad f(1)=-2,\qquad f(2)=2.$$

Thus the absolute minimum is -2, attained at $x=-2$ and $x=1$; the absolute maximum is 2, attained at $x=-1$ and $x=2$.

WORKED EXAMPLE

An endpoint may also be the minimum

For $f(x) = x + 1/x$ on $[1/2, 4]$,

$$f'(x) = 1 - \frac{1}{x^2} = \frac{(x-1)(x+1)}{x^2}.$$

The interior critical point is $x = 1$, and

$$f(1/2) = 5/2, \qquad f(1) = 2, \qquad f(4) = 17/4.$$

The absolute minimum is 2 at $x = 1$ and the absolute maximum is $17/4$ at $x = 4$.

On the larger interval $(0, \infty)$, the global minimum is still 2 because $(\sqrt{x} - 1/\sqrt{x})^2 \geq 0$. There is no global maximum since $f(x) \to \infty$ as $x \to \infty$.

COMMON TRAP

On open intervals, distinguish an attained value from a limiting value

For $x, y > 0$ with $x + y = 6$, $x^3 + y^3 = 18(x-3)^2 + 54$. The minimum 54 occurs at $x = y = 3$. The value 216 is only approached as an endpoint is approached, so there is no attained maximum.

LESSON 08 / 12

Convexity is about tangents below and secants above

On an interval, a twice-differentiable function with $f'' \geq 0$ is convex; one with $f'' \leq 0$ is concave. For a differentiable convex function,

$$f(x) \geq f(a) + f'(a)(x - a),$$

so every tangent lies below the graph. A **point of inflection** is a point where the concavity changes sign, not merely a point where f'' is zero.

WORKED EXAMPLE

A cubic with one inflection point

For $f(x) = x^3 - 3x$,

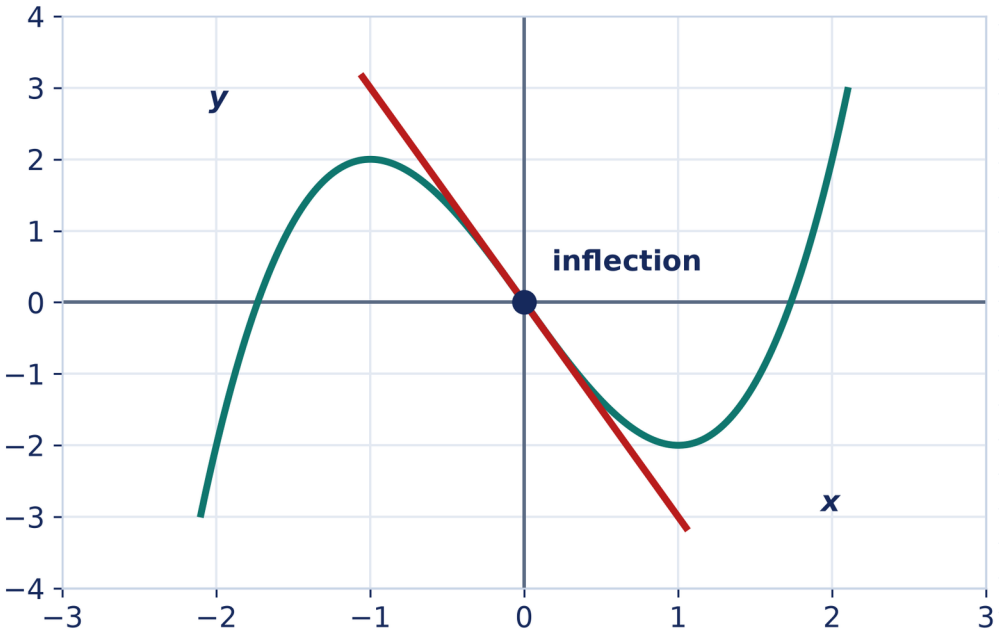
$$f''(x) = 6x.$$

It is concave on $(-\infty, 0)$ and convex on $(0, \infty)$, so $(0, 0)$ is an inflection point. The tangent there is $y = -3x$, and

$$f(x) - (-3x) = x^3,$$

which changes sign at 0: the graph crosses its tangent.

At a cubic inflection point the curve crosses its tangent



READ THE FIGURE

The teal cubic is concave left of the origin and convex right of it. Its red tangent y=-3x lies above on the left and below on the right.

For $g(x) = x^3 - 3x^2$, $g''(x) = 6(x - 1)$, so g is concave before 1 and convex after 1; $(1, -2)$ is its inflection point.

WORKED EXAMPLE

A quartic has two inflection points, but x^4 has none

For $h(x) = x^4 - 4x^2$,

$$h''(x) = 12x^2 - 8 = 12\left(x^2 - \tfrac{2}{3}\right).$$

The sign changes at $x = \pm\sqrt{2/3}$, where $h = -20/9$. Both are inflection points. The companion paper's printed $(\pm 1/\sqrt{3}, -11/9)$ do not satisfy $12x^2 - 8 = 0$; the values here correct that source defect.

For x^4, $f''(x) = 12x^2$ is zero at 0 but nonnegative on both sides. The origin is a minimum, not an inflection point.

WORKED EXAMPLE

The supporting line of x^2 at a=3

For $f(x) = x^2$, $f(3) = 9$ and $f'(3) = 6$, so the supporting line is

$$y = 9 + 6(x - 3) = 6x - 9.$$

Convexity gives $x^2 \geq 6x - 9$ for every real x. Equality occurs only at $x = 3$, as the independent identity $x^2 - (6x - 9) = (x - 3)^2$ confirms.

COMMON TRAP

Always test a sign change

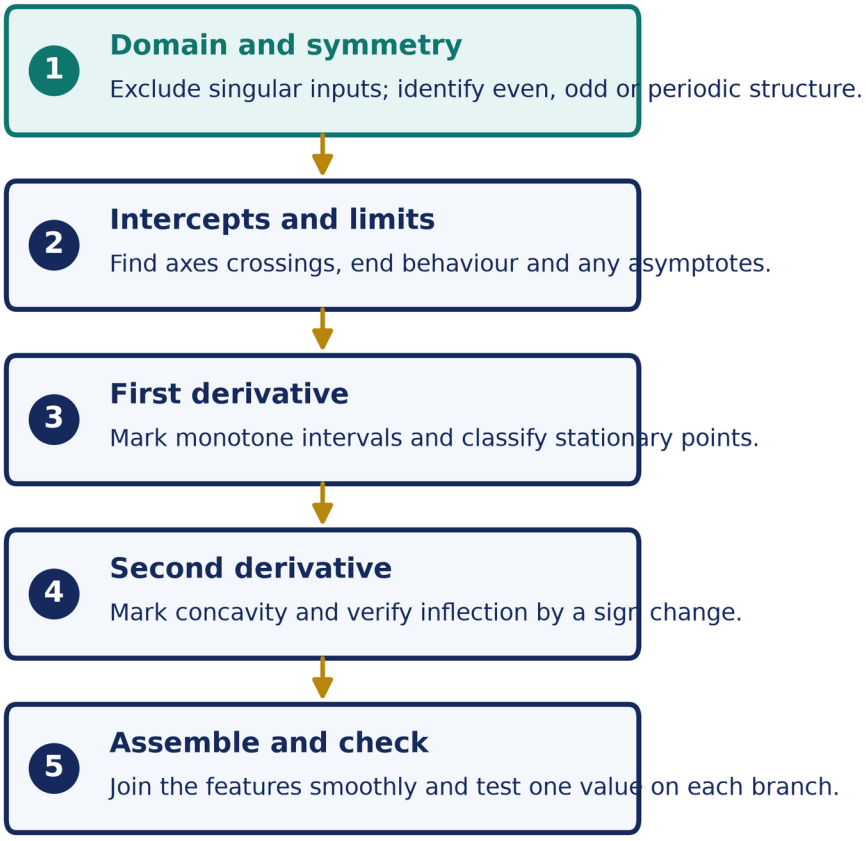
The condition $f''(c) = 0$ is neither necessary in every nonsmooth setting nor sufficient by itself. In the usual twice-differentiable setting, it is only a screening condition for a possible inflection.

LESSON 09 / 12

Combine features instead of plotting isolated points

A reliable sketch records the domain, symmetry, intercepts, asymptotic behaviour, monotone intervals, local extrema, concavity and inflection points. Each feature should agree with the others: a local maximum on a concave cap, for example, cannot sit in an interval where the second derivative is positive.

A curve-sketching checklist turns derivatives into shape



READ THE FIGURE

The graph is the final consistency check of the algebra, not a replacement for it.

WORKED EXAMPLE

A complete feature list for y=x^4-4x^2

The function is even, so its graph is symmetric about the y-axis. Its intercepts satisfy

$$x^2(x^2-4)=0,$$

giving $(0,0)$, $(-2,0)$ and $(2,0)$. The origin is a double root, so the graph touches the x-axis there rather than crossing it.

From $f'(x)=4x(x^2-2)$, the function decreases on $(-\infty,-\sqrt{2})$ and $(0,\sqrt{2})$, and increases on $(-\sqrt{2},0)$ and $(\sqrt{2},\infty)$. Its local minimum value is -4 at $x=\pm\sqrt{2}$, and its local maximum value is 0 at $x=0$.

From $f''(x)=12x^2-8$, it is convex outside $\pm\sqrt{2/3}$ and concave between them. The inflection points are

$$\left(-\sqrt{\tfrac{2}{3}},-\tfrac{20}{9}\right),\qquad \left(\sqrt{\tfrac{2}{3}},-\tfrac{20}{9}\right).$$

Finally $f(x)\to\infty$ as $x\to\pm\infty$, so both outer branches rise.

Translate the feature list into a shape

Start at the upper-left branch, descend to the left minimum, rise to the origin, descend to the right minimum, and then rise to the upper-right. The tangent is horizontal at all three turning points; at the origin the horizontal tangent is also the x-axis.

COMMON TRAP

An intercept's multiplicity affects the sketch

At $x=0$ in this example the factor x^2 has even multiplicity. The graph touches the axis. At $x=\pm2$ the factors have odd multiplicity, so the graph crosses it.

LESSON 10 / 12

Reduce to one variable, but retain the feasible interval

An optimisation model has an objective, a constraint and a domain. Differentiate the reduced objective, then check every feasible candidate. A stationary point outside the physical domain is irrelevant.

WORKED EXAMPLE

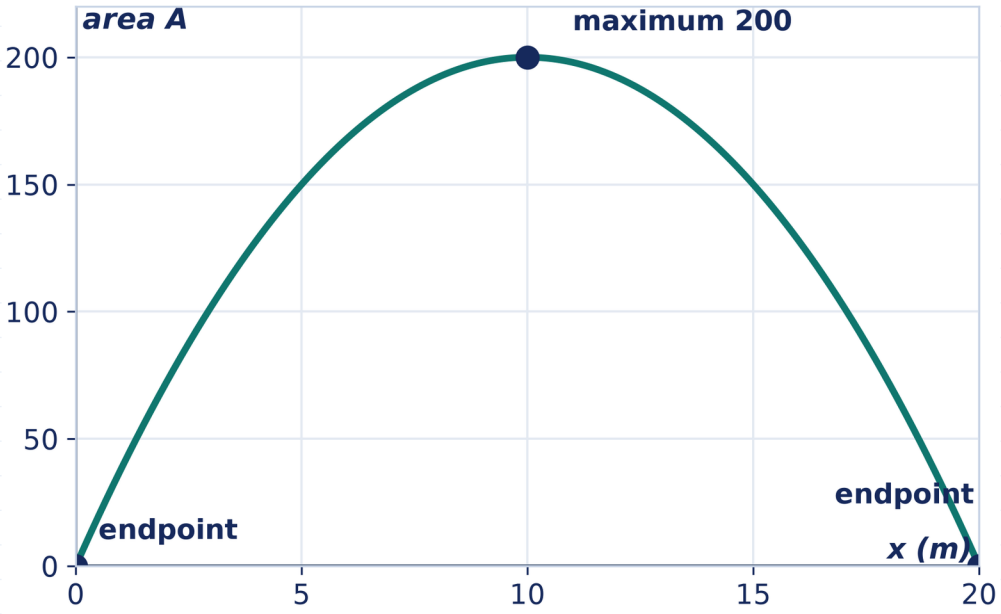
Fencing with a wall: the optimum is interior

Let x be each side perpendicular to the wall and y the parallel side. With 40 m of fence,

$$2x + y = 40, \qquad A = x(40 - 2x), \qquad 0 \le x \le 20.$$

Since $A'(x) = 40 - 4x$, the only interior stationary point is $x = 10$. The endpoint areas are zero, while $A(10) = 200$. Thus the maximum area is 200 m^2, with dimensions 10 m by 20 m.

The feasible fencing area is a downward parabola



READ THE FIGURE

Here x is the side perpendicular to the wall. The vertex x=10 is feasible and gives y=20; both zero-area endpoints remain in the candidate comparison.

WORKED EXAMPLE

An open box: the second derivative and endpoints agree

From a 12 cm by 12 cm sheet, cut squares of side x from the corners. Then

$$V(x) = x(12 - 2x)^2, \qquad 0 \le x \le 6.$$

Differentiate and factor:

$$V'(x) = (12 - 2x)(12 - 6x) = 12(6 - x)(2 - x).$$

The feasible interior stationary point is $x = 2$, where $V = 128$ cm^3 and $V''(2) = -48 < 0$. Since $V(0) = V(6) = 0$, this is the absolute maximum.

WORKED EXAMPLE

Product with fixed sum; area with fixed perimeter

For positive x, y with $x + y = 20$,

$$P = x(20 - x) = 100 - (x - 10)^2,$$

so the maximum product is 100 at $x = y = 10$. The same calculation with sum 10 gives maximum 25 at $x = y = 5$.

For a rectangle of fixed perimeter p, write $l + b = p/2$ and

$$A = l\left(\frac{p}{2} - l\right) = \frac{p^2}{16} - \left(l - \frac{p}{4}\right)^2.$$

The greatest area is $p^2/16$, attained by the square with $l = b = p/4$. In both models the completed square proves the global conclusion.

WORKED EXAMPLE

Three positive factors with sum 1

For $x + y + z = 1$, fix x. Since $y + z = 1 - x$, the product yz is at most $(1 - x)^2/4$. Hence

$$xyz \le B(x) = \frac{x(1 - x)^2}{4}, \qquad 0 < x < 1.$$

Now $4B'(x) = (1 - x)(1 - 3x)$, so B increases up to $x = 1/3$ and decreases afterwards. Equality in the first step requires $y = z$. Thus the maximum is $1/27$ at $x = y = z = 1/3$.

COMMON TRAP

Do not drop the constraints after eliminating a variable

The expression $x(12 - 2x)^2$ makes algebraic sense for $x > 6$, but those values produce a negative base length and are not box models. The feasible interval is part of the answer.

LESSON 11 / 12

Parametrise the constraint before differentiating

For a rectangle inscribed in $x^2/a^2 + y^2/b^2 = 1$, let the first-quadrant corner be $(a\cos t, b\sin t)$ with $0 \le t \le \pi/2$. The full area is

$$A(t) = 4ab\sin t\cos t = 2ab\sin 2t.$$

The maximum is $2ab$ at $t = \pi/4$, with corner $(a/\sqrt{2}, b/\sqrt{2})$. Equivalently, with corner (x, y),

$$A = 4bx\sqrt{1 - \tfrac{x^2}{a^2}},$$

and A^2 can be maximised as a quadratic in x^2.

WORKED EXAMPLE

The classic window model has a feasible optimum

A rectangle of height h and width $2r$ is topped by a semicircle of radius r. Its perimeter is

$$2h + 2r + \pi r = 10,$$

so $h = 5 - r(1 + \pi/2)$ and feasibility requires $h \ge 0$, hence

$$0 \le r \le \tfrac{10}{2+\pi}.$$

Its area, including the semicircle, is

$$A = 2rh + \tfrac{1}{2}\pi r^2 = 10r - (2 + \tfrac{\pi}{2})r^2.$$

Therefore

$$A'(r) = 10 - (4 + \pi)r, \qquad r_* = \tfrac{10}{4+\pi}.$$

At this point $h = 5 - r_*(1 + \pi/2) = r_*$, so the optimum is feasible. Since $A'' = -(4 + \pi) < 0$ and the endpoint areas are smaller,

$$A_{\max} = \tfrac{50}{4+\pi}, \qquad h = r = \tfrac{10}{4+\pi}.$$

Three optimisation models, each with its domain retained

Model	Feasible reduction	Verified optimum
Open box	0<=x<=6	x=2, V=128
Ellipse rectangle	x=a cos t, y=b sin t	A_max=2ab
Classic window	0<=r<=10/(2+pi)	r=h=10/(4+pi)

READ THE FIGURE
The window calculation uses the semicircle arc in the perimeter and its area in the objective. It does not cancel the semicircle's area term.

Audit the source algebra, not only the numerical answer

A tempting incorrect step is to claim that the rectangular area $2rh$ and semicircle area $\pi r^2/2$ cancel. They do not. After substituting the perimeter relation, the correct coefficient of r^2 is $-(2+\pi/2)$, and the resulting vertex satisfies the height constraint.

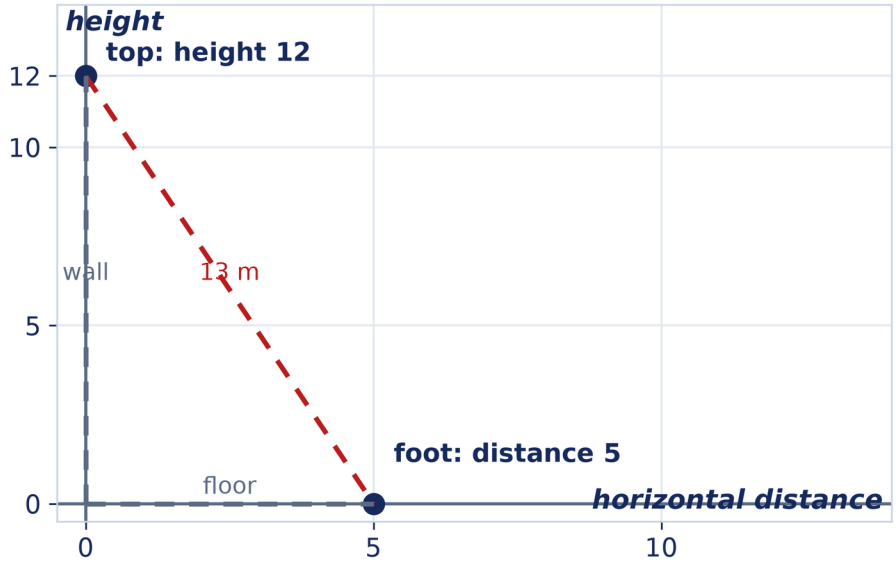
COMMON TRAP
A defect is corrected by returning to the model
The canonical source's assertion that this window optimum is infeasible is wrong. Recompute both the reduced objective and the feasible interval; the independent check in this chapter also verifies the optimum by a completed square.

LESSON 12 / 12

This lab separates the geometric and numerical ideas that can be hidden by a formula. Every diagram is generated from an editable mathematical expression; the companion checks test its anchors independently.

Lab 1 — A ladder's dimensions satisfy a circle constraint

The physical 5-12-13 ladder state gives a Pythagorean constraint



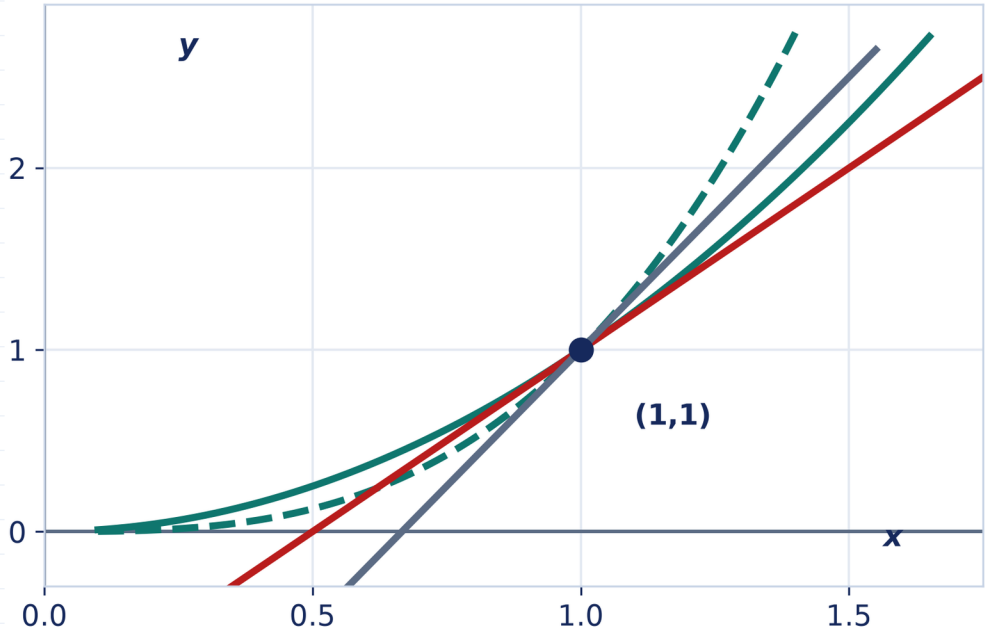
READ THE FIGURE

In this physical diagram the foot distance is x=5 and the top height is y=12, so the ladder length is sqrt(x^2+y^2)=13. Differentiate x^2+y^2=169 to obtain the time rates.

The equation $x^2 + y^2 = 169$ is a ***position constraint***, not a graph of position against time. The drawing uses physical wall/floor coordinates; x and y are the two distances. Time enters only after differentiating the constraint, which prevents confusing the geometric slope with the requested time rate.

Lab 2 — Angle between curves is angle between tangents

Two curves and their tangents at the common point (1,1)

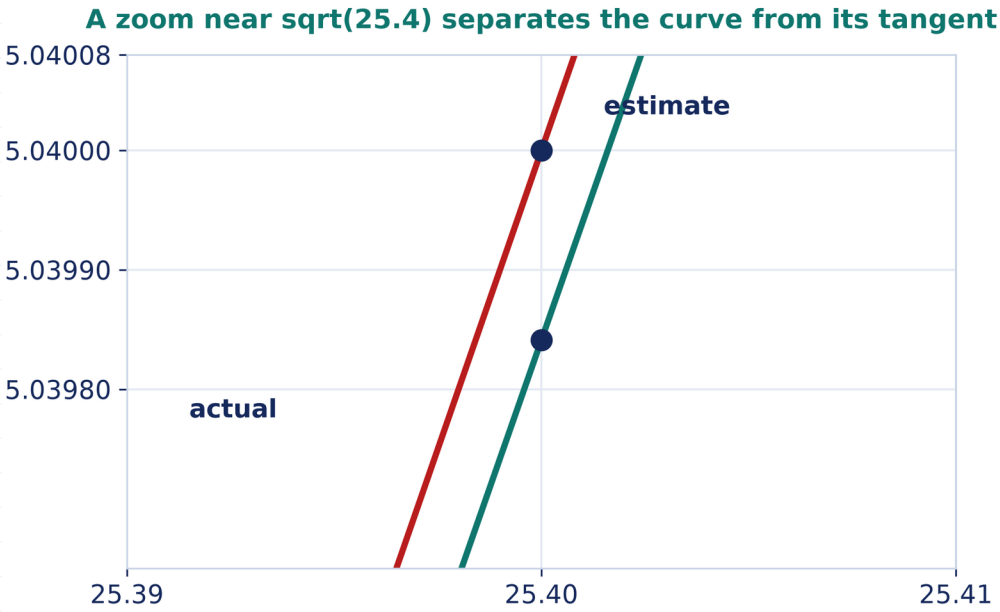


READ THE FIGURE

The teal solid parabola and dashed cubic have tangent slopes 2 and 3 at (1,1). The red and slate tangent lines make an acute angle arctan(1/7), about 8.13 degrees.

To reproduce the angle, use $\tan\theta = |(3-2)/(1+6)| = 1/7$. The tangent lines—not the secants and not the angle between radius vectors—carry the local directions of the curves.

Lab 3 — Linear approximation has a visible error under magnification



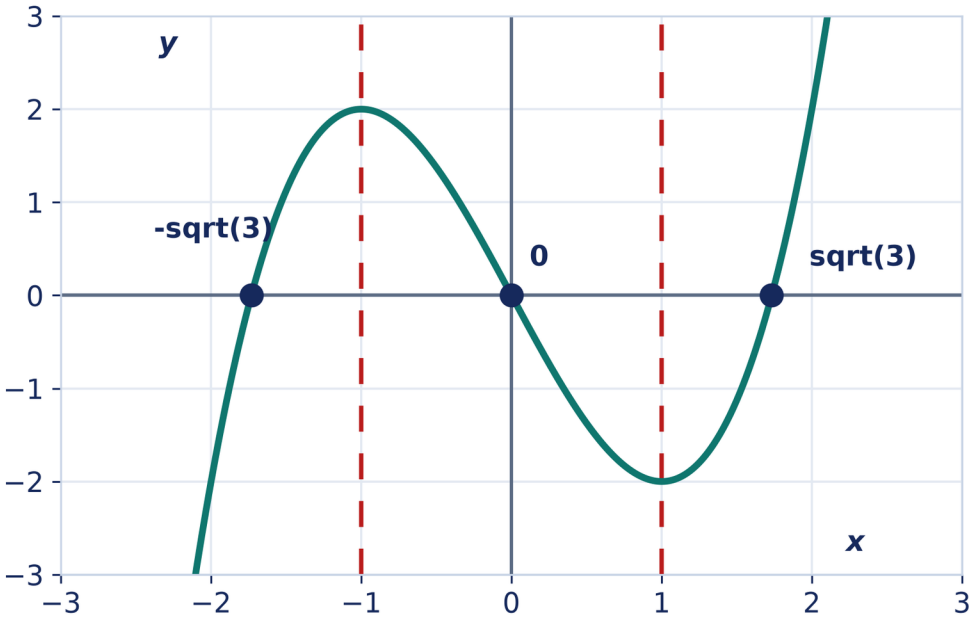
READ THE FIGURE

The red tangent at x=25 gives the estimate 5.04; the teal square-root curve gives 5.0398412677.... The vertical gap is about 0.0001587, invisible at ordinary graph scale.

For $f(x)=\sqrt{x}$, $f''(x)=-1/(4x^{3/2})<0$. Concavity explains why the tangent estimate is above the actual value. The next-order term is approximately $f''(25)(0.4)^2/2=-0.00016$, matching the displayed error.

Lab 4 — Three monotone branches give exactly three roots

Each monotone branch of the cubic crosses the x-axis once



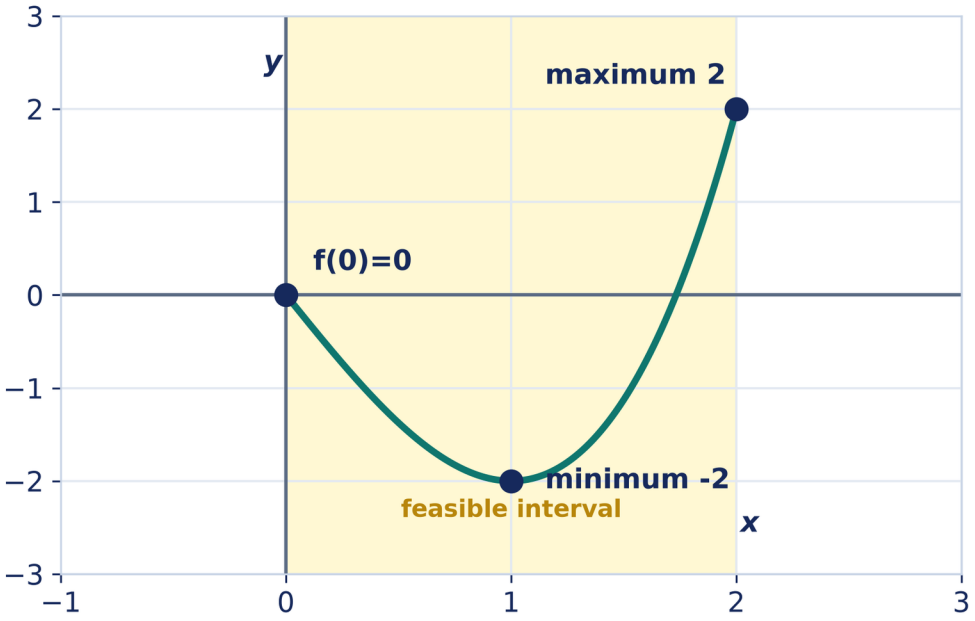
READ THE FIGURE

The red dashed lines x=-1 and x=1 separate three strictly monotone branches. The signs at the turning values 2 and -2, together with the end behaviour, give one crossing per branch.

The graph illustrates the proof but does not replace it. The algebraic argument uses IVT for existence and strict monotonicity for uniqueness, then verifies the roots $-\sqrt{3}, 0, \sqrt{3}$ exactly.

Lab 5 — Absolute extrema include both endpoints

On [0,2], the cubic's maximum occurs at the right endpoint

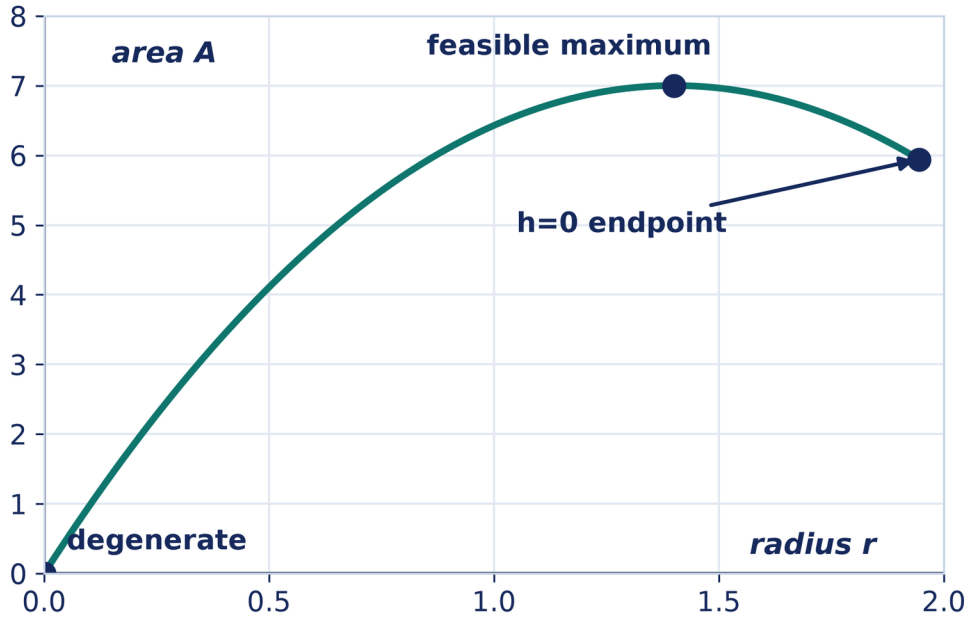


READ THE FIGURE
For f(x)=x^3-3x on [0,2], the candidate values are f(0)=0, f(1)=-2 and f(2)=2. Comparing values—not slopes—selects the absolute extrema.

The local maximum at $x=-1$ is outside this interval and must not enter the candidate list. Conversely, the endpoint $x=2$ must enter even though the derivative there is not zero.

Lab 6 — The corrected window optimum is feasible

The window area vertex lies before the zero-height endpoint



READ THE FIGURE

The feasible interval ends at r_max=10/(2+pi), about 1.9449, where h=0 and the semicircle area is 5.9419. The vertex r*=10/(4+pi), about 1.4002, gives the larger value A_max=50/(4+pi), about 7.0012.

At $r = r_*$, the perimeter relation gives $h = r_* > 0$. At $r = r_{\max}$, the height is zero and the window degenerates to a semicircle. The completed-square identity in the verification script independently proves that the interior vertex gives the global maximum.

LESSON 01 / 05

One tangent can prove a global inequality

If f is differentiable and convex, then for every a and x in its interval,

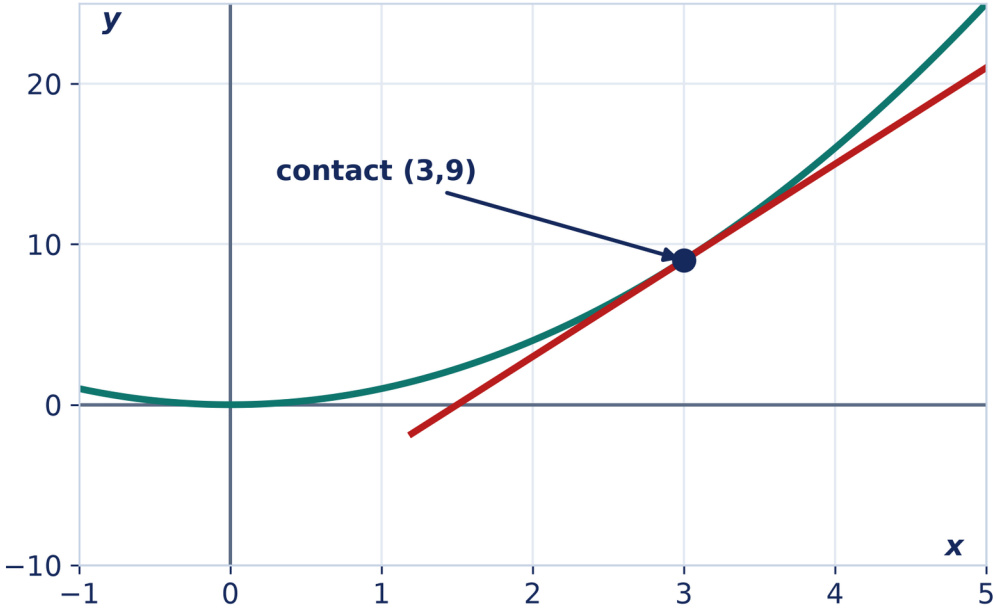
$$f(x) \geq f(a) + f'(a)(x - a).$$

For $f(x) = x^2$ at $a = 3$, the supporting line is

$$y = 9 + 6(x - 3) = 6x - 9.$$

Thus $x^2 \geq 6x - 9$, with equality only at $x = 3$. Independently, the difference is $(x - 3)^2 \geq 0$.

A convex parabola lies above its supporting line



READ THE FIGURE
The red supporting line y=6x-9 touches the teal graph y=x^2 at x=3 and stays below it everywhere. Their difference is (x-3)^2.

Jensen follows by averaging supporting lines

Let $\lambda_i \geq 0$ and $\sum_{i=1}^{n} \lambda_i = 1$, and put $\bar{x} = \sum_{i=1}^{n} \lambda_i x_i$. Apply the supporting-line inequality at $\bar{x}$ to each x_i, multiply by λ_i, and add. The first-order terms cancel because $\sum \lambda_i (x_i - \bar{x}) = 0$. Hence

$$f(\bar{x}) \leq \sum_{i=1}^{n} \lambda_i f(x_i).$$

For strictly convex f, equality requires all the x_i with positive weights to coincide.

WORKED EXAMPLE

Jensen for the exponential

Since $D^2(e^x) = e^x > 0$, *for positive* a, b, c,

$$\frac{e^a + e^b + e^c}{3} \geq e^{(a+b+c)/3},$$

with equality exactly when $a = b = c$. *At* $(1, 2, 3)$, *the left side is approximately* 10.0643 *and the right side is* $e^2 \approx 7.3891$, *so the inequality is strict.*

Convexity of a difference can locate its global minimum

WORKED EXAMPLE

Prove cos x >= 1-x^2/2 for every real x

Let

$$g(x) = \cos x - 1 + \frac{x^2}{2}.$$

Then $g'(x) = x - \sin x$ *and* $g''(x) = 1 - \cos x \geq 0$. *Thus* g' *is increasing. Since* $g'(0) = 0$, *we have* $g' < 0$ *for* $x < 0$ *and* $g' > 0$ *for* $x > 0$. *Therefore* g *decreases up to* 0 *and increases afterwards, so*

$$g(x) \geq g(0) = 0.$$

Equality holds only at $x = 0$. *The argument uses monotonicity of* g', *not an unsupported claim that* $g'' \geq 0$ *alone makes* g *nonnegative.*

LESSON 02 / 05

Young's inequality is weighted AM–GM in exponential form

For $a, b \geq 0$ and conjugate exponents $p, q > 1$ with $1/p + 1/q = 1$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

For positive a, b, weighted AM–GM with weights $1/p$ and $1/q$ gives

$$\frac{a^p}{p} + \frac{b^q}{q} \geq (a^p)^{1/p}(b^q)^{1/q} = ab.$$

Equality holds exactly when $a^p = b^q$, equivalently $a^{p-1} = b$; the boundary equality cases are checked directly.

WORKED EXAMPLE

Young with p=q=2 proves two-variable AM–GM

For $a, b > 0$, apply Young to $\sqrt{a}$ and $\sqrt{b}$:

$$\sqrt{ab} \leq \frac{(\sqrt{a})^2 + (\sqrt{b})^2}{2} = \frac{a+b}{2}.$$

Equality holds exactly when $\sqrt{a} = \sqrt{b}$, hence when $a = b$.

AM–GM for three positive numbers from concavity

Since $(\ln x)'' = -1/x^2 < 0$, Jensen for the concave logarithm reverses the inequality:

$$\frac{\ln x + \ln y + \ln z}{3} \leq \ln\left(\frac{x+y+z}{3}\right).$$

Exponentiating gives

$$\sqrt[3]{xyz} \leq \frac{x+y+z}{3},$$

with equality exactly when $x = y = z$. This proves Q36. The same concave-log Jensen argument with weights $1/p$ and $1/q$ proves the weighted AM–GM step used above.

Cauchy–Schwarz starts with a direct square

For all real a, b,

$$a^2 + b^2 - 2ab = (a-b)^2 \geq 0,$$

so $a^2 + b^2 \geq 2ab$, with equality exactly when $a = b$. The source's alternative expression containing $\sqrt{a^2b^2}$ equals $|ab|$, not ab; it is invalid when $ab < 0$. The displayed square is valid for all real numbers.

Summing the two-variable inequality with suitable normalisation leads to Cauchy–Schwarz. In Engel form, for real a_i and positive x_i,

$$\left(\sum_{i=1}^{n} a_i\right)^2 = \left(\sum_{i=1}^{n} \frac{a_i}{\sqrt{x_i}}\sqrt{x_i}\right)^2 \le \left(\sum_{i=1}^{n} \frac{a_i^2}{x_i}\right)\left(\sum_{i=1}^{n} x_i\right).$$

Thus $\sum a_i^2/x_i \ge (\sum a_i)^2/\sum x_i$.

The inequality family has distinct hypotheses and equality cases

Result	Main hypotheses	Equality
Young	1/p+1/q=1	a^p=b^q
2-variable AM-GM	a,b>0	a=b
3-variable AM-GM	x,y,z>0	x=y=z
Square bound	a,b real	a=b
Cauchy Engel	x_i>0	a_i proportional to x_i

READ THE FIGURE

Use the signed square for all real a,b. Use positive variables when logarithms, geometric means or Engel denominators require them.

LESSON 03 / 05

Differentiate the family with respect to its parameter

For $a > 0$ and $m \neq 0$, consider

$$F(x, y, m) = y - mx - \frac{a}{m} = 0.$$

An envelope point satisfies both $F = 0$ and $\partial F/\partial m = 0$:

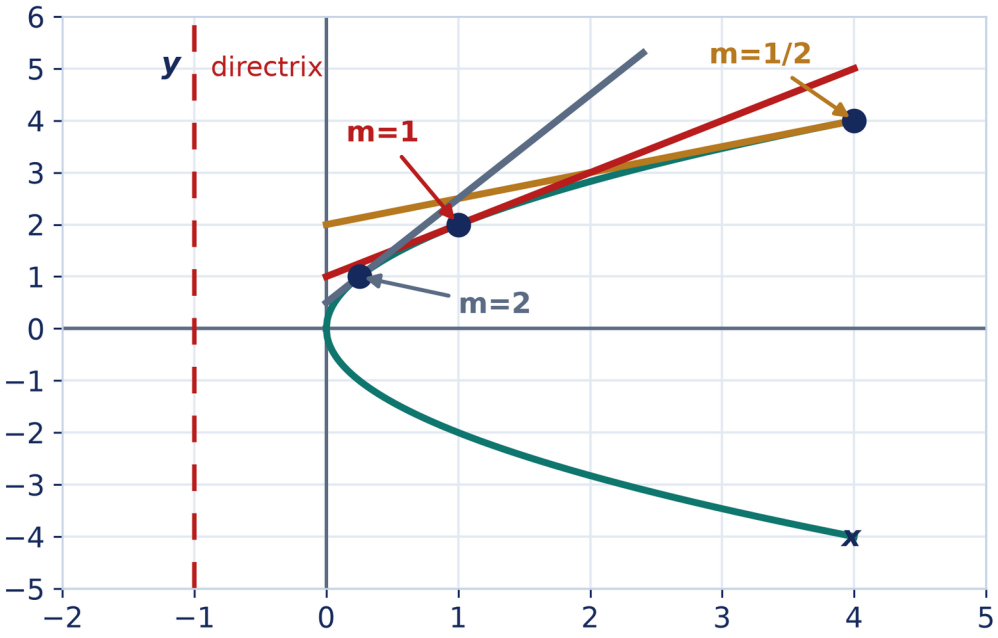
$$-x + \frac{a}{m^2} = 0 \quad \Longrightarrow \quad x = \frac{a}{m^2}.$$

Substitution into the line gives $y = 2a/m$. Therefore the contact point is

$$\left(\frac{a}{m^2}, \frac{2a}{m}\right),$$

and eliminating m yields $y^2 = 4ax$. Thus every line $y = mx + a/m$ touches this parabola at the stated point.

A family of lines envelops the parabola y^2=4x



READ THE FIGURE
For a=1, the gold, red and slate lines have m=1/2,1,2 and touch at the marked points. The red dashed line x=-1 is the directrix used in the perpendicular-tangent theorem.

Perpendicular tangents meet on the directrix

A tangent perpendicular to $y = mx + a/m$ has slope $-1/m$. Replacing m by $-1/m$ in the family gives

$$y = -\frac{x}{m} - am.$$

At the intersection of the two tangents,

$$mx + \frac{a}{m} = -\frac{x}{m} - am,$$

so

$$(m + \tfrac{1}{m})(x + a) = 0.$$

Since $m + 1/m \neq 0$ for real m, the intersection has $x = -a$, the directrix of $y^2 = 4ax$.

COMMON TRAP

Envelope differentiation uses the parameter, not x or y

The equation $\partial F/\partial m = 0$ is taken while x and y are held fixed. It supplies the second equation needed to locate the envelope and its contact point.

LESSON 04 / 05

Perpendicular ellipse tangents determine a concentric circle

For the ellipse $x^2/a^2 + y^2/b^2 = 1$, a nonvertical tangent of slope m can be written

$$y = mx \pm \sqrt{a^2m^2 + b^2}.$$

If it passes through $P = (x, y)$, rearranging and squaring gives

$$(x^2 - a^2)m^2 - 2xym + (y^2 - b^2) = 0.$$

The two tangent slopes from P are perpendicular exactly when their product is -1. By the product of roots,

$$\frac{y^2 - b^2}{x^2 - a^2} = -1 \quad \Longrightarrow \quad x^2 + y^2 = a^2 + b^2.$$

This is the **director circle**. Vertical or horizontal tangent cases satisfy the same locus directly.

WORKED EXAMPLE

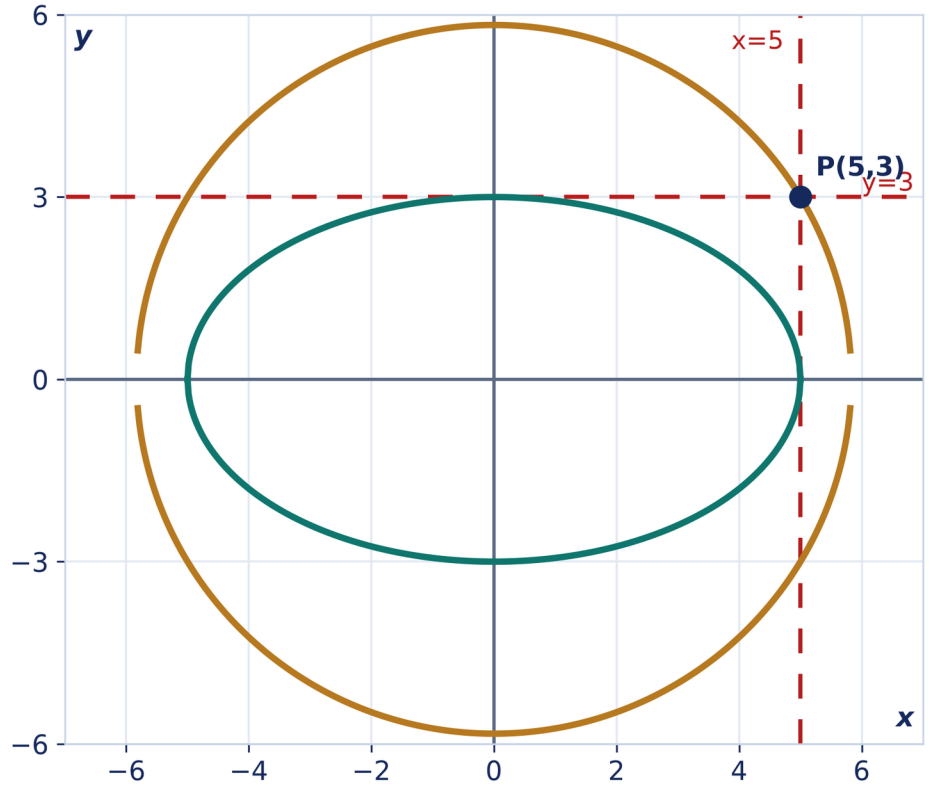
The 5-3 ellipse and the circle special case

For $x^2/25 + y^2/9 = 1$, perpendicular tangents meet at P with

$$|OP|^2 = 25 + 9 = 34, \qquad |OP| = \sqrt{34} \approx 5.831.$$

When $a = b = R$, the ellipse is a circle and the director circle has radius $\sqrt{2}R$. Thus perpendicular tangents to $x^2 + y^2 = R^2$ meet at distance $\sqrt{2}R$ from the centre.

Perpendicular tangents to the ellipse meet on its director circle



READ THE FIGURE
The teal ellipse has perpendicular tangents x=5 and y=3 at (5,0) and (0,3). They meet at P=(5,3), which lies on the gold director circle x^2+y^2=34.

Lagrange multipliers encode a constrained stationary point

To extremise f subject to $g=0$, a regular interior extremum satisfies $\nabla f=\lambda\nabla g$. For

$$f=x^2+y^2+z^2, \qquad g=x+y+z-3=0,$$

this gives $2x=2y=2z=\lambda$. Hence $x=y=z$, and the constraint gives $x=y=z=1$. The candidate value is 3.

A global proof is immediate:

$$x^2+y^2+z^2-3=(x-1)^2+(y-1)^2+(z-1)^2\geq 0$$

when $x+y+z=3$. Thus the minimum is 3 at $(1,1,1)$, not merely a stationary candidate.

COMMON TRAP
Multipliers find candidates; a separate argument establishes global extrema
Check regularity, boundaries and attainment. Here the completed squares supply the missing global comparison in one line.

LESSON 05 / 05

Symmetry suggests equal variables; monotonicity in the product proves it

For positive x, y with fixed sum t, put $s = xy$. Then

$$0 < s \le \frac{t^2}{4},$$

with equality at $x = y = t/2$. A symmetric expression rewritten in terms of t and s is optimised by checking whether it increases or decreases in s.

WORKED EXAMPLE

Two minima with x+y=10

Since $s = xy \le 25$,

$$\frac{1}{x} + \frac{1}{y} = \frac{10}{s} \ge \frac{10}{25} = \frac{2}{5},$$

and

$$x^2 + y^2 = 100 - 2s \ge 100 - 50 = 50.$$

Both minima occur at $x = y = 5$.

For positive x, y, z with $x + y + z = 1$, AM–GM gives

$$xyz \le \left(\frac{x+y+z}{3}\right)^3 = \frac{1}{27},$$

with equality at $x = y = z = 1/3$. A one-variable alternative fixes x and uses $yz \le (1-x)^2/4$, then maximises $x(1-x)^2/4$ by differentiation.

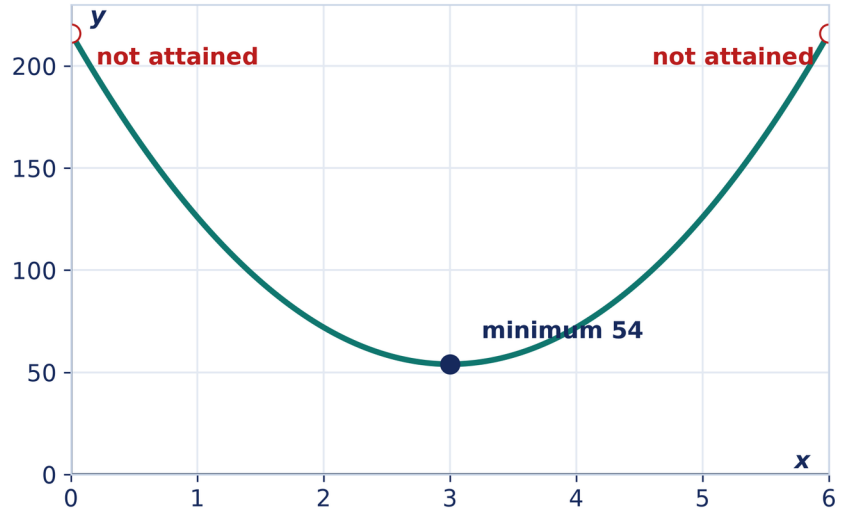
The symmetric cubic is a counterexample to blind equalising

For positive x, y with $x + y = 6$,

$$x^3 + y^3 = 216 - 18xy = 18(x-3)^2 + 54.$$

Since $xy \le 9$, the attained minimum is 54 at $x = y = 3$. But as $x \to 0^+$ and $y \to 6$, the value approaches 216 from below. Therefore the **supremum is 216 and no maximum is attained** on the positive domain.

The symmetric cubic has an interior minimum but no attained maximum



READ THE FIGURE
The graph shows x^3+y^3 after substituting y=6-x. Hollow red endpoints mark limits outside the positive domain; the interior point (3,54) is the attained minimum.

As a numerical check, at $(x,y)=(0.1,5.9)$ the value is $0.001+205.379=205.38$, close to but below 216. This corrects the companion solution's printed check value 205.8.

Equal variables minimise a sum of squares under a plane constraint

For positive x,y,z with $x+y+z=1$,

$$\left(x-\tfrac{1}{3}\right)^2+\left(y-\tfrac{1}{3}\right)^2+\left(z-\tfrac{1}{3}\right)^2=x^2+y^2+z^2-\tfrac{1}{3}.$$

Hence

$$x^2+y^2+z^2\geq\tfrac{1}{3},$$

with equality exactly when $x=y=z=1/3$. The same completed-square argument proves the Lagrange example in Extension IV.

COMMON TRAP
Always identify whether the objective increases or decreases in s
Equal variables maximise xy and minimise x^2+y^2 for a fixed positive sum, but they minimise rather than maximise x^3+y^3. Symmetry alone does not choose the type of extremum.